

Reproduction I

SPECIES MUST REPRODUCE TO SURVIVE.

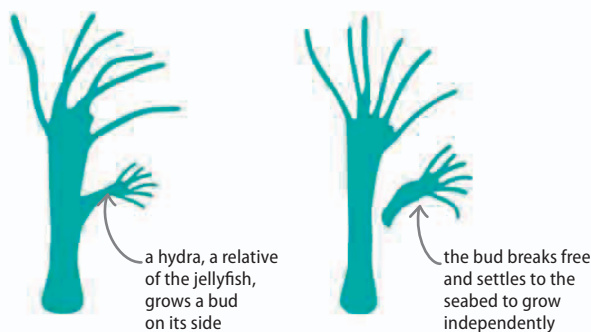
Reproduction is the main purpose of the natural world. Living things grow, feed, and survive in order to reproduce and makes copies of themselves.

Asexual reproduction

When a single organism makes an exact copy of itself, the process is called asexual reproduction. The copy is genetically identical, a clone of the parent. Asexual reproduction can be useful for populating new habitats very quickly. However, because all the offspring are identical, a disease or other problem that affects one of them is likely to affect all the others, too.

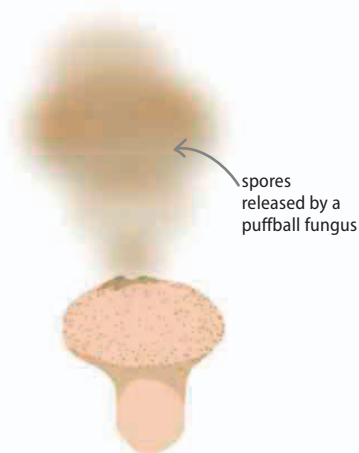
▽ Budding

The most basic form of reproduction is budding, in which a section of the parent breaks off, forming an independent individual. Many single-celled organisms reproduce by budding.



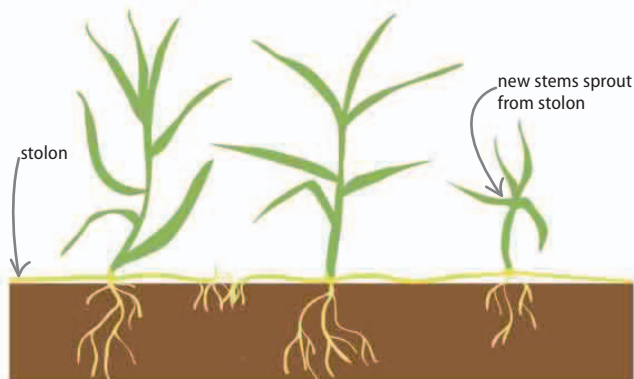
▷ Sporogenesis

Fungi, primitive plants (such as ferns and moss), and even some parasitic worms reproduce by releasing hardy spores. These are tiny balls of cells, which can grow into new individuals.



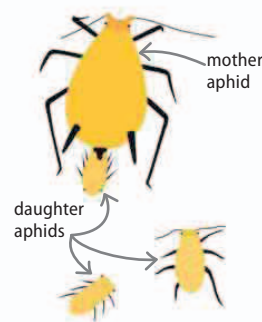
▽ Vegetative reproduction

Some plants send out side roots (called runners) or stems (stolons), that sprout daughter plants nearby. When the daughter plant is established, the connection with the parent breaks.



▷ Parthenogenesis

Parthenogenesis is a form of reproduction in which animals produce young without mating. Some female aphids give birth to daughters that are identical to themselves in every way except size.



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◀ 25 Cell division

Human reproduction 72–73 ▶

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New Mexico whiptail lizards are all asexual, but all females must “mock mate” with each other before laying eggs.